

Critical Review

Accessibility, equity and health care: review and research directions for transport geographers



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ABSTRACT

Research at the confluence of accessibility, equity and health is flourishing. And yet, there is only slow and modest progress in terms of improving the spatial and temporal accuracy of measuring accessibility and equity of accessibility to health care services. This paper critically reviews the latest methodological and empirical research developments and trends in this area through a transport geography lens. More specifically, this paper discusses recent accomplishments in terms of modeling accessibility and provides a systematic and comprehensive literature review of their application in empirical studies of health care delivery. Based on this literature review a research agenda is put forward, identifying knowledge gaps that transport researchers can help to fill. These knowledge gaps pertain to the need for more spatially disaggregated, individualized and temporally-aware accessibility metrics, more sophisticated geocomputational tools to operationalize such metrics and improved measurement of equity considerations in empirical research.

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1. Introduction

Health care access is concerned with the processes surrounding the entry of persons and population groups into the health care delivery system. It is a multi-faceted concept that involves five dimensions including affordability (i.e. costs of health care utilization), acceptability (i.e. health service compliance and satisfaction), availability (i.e. adequacy of health service provision), geographic accessibility (i.e. travel impedance between patients and providers) and accommodation (i.e. appropriateness and suitability of health services) (Penchansky and Thomas, 1981). This review paper concentrates on one of these aspects, namely geographic accessibility, and discusses its measurement and operationalization. The scope of this review is limited to geographic accessibility of primary care, hospital care and emergency medical services as these services are seen as the most important contributors to overall population health and have been the focal point of scholarly attention in the recent past.

The motivation for a comprehensive review on this matter is threefold. First, of the aforementioned five dimensions of health care access, it is particularly geographic accessibility, also termed spatial or physical accessibility or simply accessibility, that has gained traction over the past few years, chiefly owing to the increased sophistication of geographic information systems (GIS) and availability of spatially disaggregated data on both health care supply and demand. There is now a wide recognition of the value

of GIS in, among others, mapping the spatial distribution of health care needs and utilization, monitoring and evaluating the socio-spatial repercussions of health policy actions, determining optimal health service locations and disentangling the relationships between disparities in accessibility and health outcomes (Higgs, 2004; McLafferty, 2003).

Second, adequate and equitable accessibility is a concern of growing importance to policy makers, public health reformers and practitioners alike since recent years have brought convincing evidence to the fore showing that spatial barriers between consumer (e.g., patient) and provider (e.g., general practitioner) contribute to lower health care utilization and decreased uptake of preventive services, which may in turn give rise to poorer health outcomes. Dai (2010), for example, found that women are more likely to respond positively to invitations to attend mammography screening in Detroit (Michigan) if they lived closer to existing clinics. Hiscock et al. (2008), for their part, noted an inverse relationship between travel time access and utilization of general practitioners (GPs) and pharmacies in a nation-wide study in New Zealand. Another example is Gage and Calixte (2006) who showed that accessibility is a crucial dimension of the use of maternal health care in Haiti. Much of the empirical evidence to date has accrued from rural, sparsely populated areas where people experience distance barriers resulting from diminished local availability of health care provision. But the 'health accessibility lens' has also been applied to deprived (e.g., low-income) individuals

and communities who tend to have a poorer health status and fewer mobility resources for accessing health services relative to their high-income counterparts (Bissonnette et al., 2012; Kawakami et al., 2011).

Third, while measuring accessibility and equity in accessibility of health services is of straightforward importance in developing countries where spatial coverage problems are often obvious and acute, it also becomes an extremely apposite exercise in the developed world in light of the radical demographic changes taking place in contemporary Western societies. A steady decline in fertility rate over the past half century in combination with increased longevity is engendering a dramatic shift in the population distribution of many industrialized countries towards older ages (Lutz et al., 2008). These emerging cohort effects pose serious challenges for efficiently allocating the available health resources among the population in an equitable manner, especially in a climate of budgetary pressure (Loo et al., 2012). Hence, a realistic portrayal of health care accessibility is key to supporting policy makers in taking judicious rationing decisions that affect the spatial planning of health resources in order to maintain adequate care for a growing and more demanding yet less mobile older, age cohort.

This review paper highlights various methodological and conceptual issues in the measurement of social and spatial inequalities in accessibility of health care services. This review comes at a time when a growing number of voices are calling for temporally integrated perspectives on accessibility that account for behavioral and mobility aspects (Guagliardo, 2004; Kwan, 2012a; Miller, 2007; Páez et al., 2010b). While recent advancements in this active field of transport geography have pushed forward the research frontier in terms of both methodological and empirical aspects, it is argued that measurements of (equity in) accessibility in empirical health studies still tend to be conceptualized predominantly in static spatial terms, creating misleading health care landscapes and concomitant misinformed policy interventions (Dewulf et al., 2013; Hawthorne and Kwan, 2012). This review calls for more spatially disaggregated, individualized and temporally-aware accessibility metrics, increased accuracy in modeling transportation to and demand for health care services, improved measurement of equity considerations and more sophisticated geocomputational tools to operationalize the suggested improvements.

Several recent reviews on health care accessibility can be found in the scholarly literature. Many of these have laid out the advantages of GIS for health care research and its adoption by health researchers and policy makers, including McLafferty (2003), Higgs (2004), Joyce (2009) and Shaw (2012). Others, among them Yang et al. (2006), Apparicio et al. (2008), Schuurman et al. (2010) and Dewulf et al. (2013), have offered a comparative analysis of accessibility measures in particular health settings. Wang (2012) has specifically looked at distance decay and optimization models in this context. This review is different from these previous studies in that it focuses on the implications of the shortcomings of current accessibility measures for understanding equity in access to health care and also provides a more systematic investigation of empirical work on primary, hospital and emergency care. The closest predecessor of our review is perhaps the work by Guagliardo (2004). Apart from updating Guagliardo's work by discussing the accomplishments made in the last decade, this paper puts accessibility of health care into a different perspective – that is a transport geography perspective rather than a public health perspective. Furthermore, it expands the scope of Guagliardo's literature review and suggests new pathways for further exploration.

The remainder of this paper comes in four parts. The paper proceeds with an overview of how accessibility has been assessed in previous studies in the context of health care and identifies various methodological issues. Next, a literature review is conducted on the basis of a systematic literature search. From this literature

review, a research agenda is derived which offers various suggestions and avenues for future research for transport geographers. The paper concludes with a synthesizing discussion.

2. Methodological approaches to measuring accessibility of health care services

The concept of accessibility is deep-seated in transport geography and planning. In essence it refers to the degree to which transport systems enable people to reach desired activity locations. The concept helps to unravel the complex web of causalities between transport, human activity and land use. It can be considered the principal product of transport systems and the mainspring of many transport policies. At a collective level, accessibility steers regional development by generating positive externalities for social cohesion, environmental sustainability and economic growth (Gutiérrez et al., 2010; Vickerman et al., 1999). In health research, accessibility plays a central role in studies on the effects of environmental exposure on health behavior and outcome (Diez Roux and Mair, 2010; Perchoux et al., 2013), including the large body of work surrounding walkability (Leslie et al., 2007) and food deserts and swamps (Shannon, 2014). In the context of health care in particular, the concept is of major importance to understanding how the spatial distribution of health services may disproportionately bear on certain population groups.

While various advanced formulations of accessibility have been proposed in the transport literature, simple metrics such as distance and travel time to the nearest service and population-to-provider ratios (PPR) are being overwhelmingly favored among health scholars and practitioners. There are two reasons for this. First, these indicators are relatively easy to implement by public health researchers with a basic knowledge of GIS. And second, these indicators are directly interpretable in absolute units and are therefore easily communicable to and readily understood by policy makers.

Apart from these rudimentary metrics, the slightly more complex gravity models derived from social physics has also seen a fair number of applications in the health arena (Crooks and Schuurman, 2012; Henry et al., 2013; Joseph and Bantock, 1982; Khan, 1992). Gravity measures are based on the seminal work by Hansen (1959) and take the following general form:

$$A_i = \sum_j \frac{a_j}{f(c_{ij})}$$

where i and j denote the location of health demand and health supply, respectively; a represents the size or quality of health supply at j ; f is a distance or travel time decay function; and c_{ij} denotes the travel time, distance or generalized cost between i and j . Gravity models thus trade off the size and/or quality of health care facilities with travel impedance. Various subtypes of this genre exist depending on how this trade-off is achieved. In contrast to transport practice where the negative exponent of travel distance is the preferred function to capture decay, health care studies often work with piecewise decay functions consisting of constant arbitrary impedance values for a priori defined bandwidths of travel distance. This is in part because negative exponential decay parameters cannot be reliably estimated without detailed information about the evolution of utilization rates in function of distance. This information can be derived from patient registers but these are seldom available for academic work because of privacy regulations. A specific form of piecewise decay function is a rectangular function that takes the value of 1 if health supply is within predetermined distance from health demand, and 0 otherwise. This effectively makes the gravity measure equivalent to a cumulative opportunity measure, which is also known as isochrone measure, contour measure, proximity count or container index in the literature. In other words, a

cumulative opportunity measure tallies up the number of opportunities (e.g., hospital beds) within a certain cut-off distance. Cumulative opportunity counts have the advantage that they represent comparable and absolute units, whereas values resulting from gravity measures with continuous travel decay functions can only be interpreted in relative terms by normalizing the values over a particular range (Batty, 2009). A disadvantage of the approach is, however, the arbitrariness involved in the selection of an appropriate cut-off distance.

An interesting solution to this selection problem has been proposed by Páez et al. (2010b, 2013) who have used empirically informed estimates of average (straight-line) trip length (irrespective of the point of departure, purpose and mode) as cut-off distances in cumulative opportunity measures. These average trip lengths have been estimated in a multivariate spatial modeling framework with socio-economic, mobility and location characteristics as exploratory variables using Casetti's (1972) expansion method. This method allows regression coefficients to vary spatially, resulting in location-specific cut-off values for different personal profiles. While technically elegant, the use of average trip length as a proxy for "transportation burden" is not without problems from a conceptual point of view. One issue is that observations of how far people actually travel is generally a poor indicator of their willingness and opportunities to travel. For example, a person who conducts no trips other than a long commuting trip by public transport at the day of the survey is considered willing and able to search for health opportunities within a large radius around the home location, even if that person lives in a rural area and does not have access to a car. Equally, a person who is very mobile and makes a large number of short trips per day has on average a short trip distance within which to access health opportunities. Furthermore, average distance travelled varies strongly for different purposes and across modes. In addition, there are also significant day-to-day variations in average distance travelled and accessibility as a consequence of occasional deviations to repetitious regimes of space-time constraints (Neutens et al., 2012). Some of these idiosyncrasies may average out at the profile level provided that a large trip survey with pooled observation days is available within the considered study area, but the problems at the conceptual level nonetheless remain.

Another descendant of the gravity model is the two-step floating catchment area (2SFCA) method. This method deserves particular attention here given the recent flurry of applications in health care studies. As the name implies, the computation of the 2SFCA measure consists of two steps: (i) determining the population that falls within the catchment of each health care provider to compute the provider-to-population ratio, and (ii) allocating providers to populations by determining which providers fall within the catchment of each population and sum up the population-to-provider ratios obtained in the first step of the procedure. Consider the simplified example, adapted from an earlier example of Luo and Wang (2003), of a fictive study area consisting of 14 zones and 14 residents, one per zone (Fig. 1).

Two health care providers s_1 and s_2 are active in the considered area. In the first step of the procedure (Fig. 1a), a catchment of a particular distance is drawn around each provider location and all populations within this catchment are summed to compute a weighted provider-to-population ratio:

$$R_j = \frac{S_j}{\sum_{k \in \{d_{kj} \leq d_r\}} P_k} W(d_{kj})$$

where P_k is the population of zone k whose centroid falls within the catchment, S_j is the number of health care providers at location j and d_{kj} is the distance between k and j , and $W(d_{kj})$ is a distance decay function of d_{kj} . S_j is larger than 1 if multiple health care

providers are available at j and W monotonically decreases with increasing d_{kj} . In the example, R_{s_1} will be $1/5$ and R_{s_2} will be $1/4$, since 5 and 4 persons reside within the catchments of provider s_1 and s_2 , respectively. The second step of the procedure (Fig. 1b) is to draw a catchment around the population centroids and simply add up all weighted provider-to-population ratios of the providers that fall within it:

$$A_i = \sum_{j \in \{d_{ij} \leq d_r\}} R_j W(d_{ij})$$

where A_i represents the accessibility at residence location i and d_{ij} is the distance between i and j . For the individual living in zone 9, for instance, the ratio-based accessibility level A_9 equals $9/20$ ($1/5 + 1/4$).

The 2SFCA is appealing because it controls for capacity restrictions, local competition and cross-border healthcare-seeking behavior. Since its introduction by Luo and Wang (2003), the basic model has undergone various incremental enhancements, including the incorporation of variable catchment sizes (McGrail and Humphreys, 2009), distance decay effects (Luo and Qi, 2009), age adjusted health care demand (Ngu and Apparicio, 2011) and multiple transport modes (Mao and Nekorchuk, 2013). While the 2SFCA method has proven successful in identifying spatial medical deficits, at least three fundamental shortcomings can be identified. First, the 2SFCA method is a static spatial measure. It does not allow travel time, supply and demand to vary over time, and hence ignores important real-world aspects of health care provision such as consultation hours, time-dependent staffing capacities, waiting times at health care facilities, work time and work location of patients. Second, local demand for health care services is overestimated in the 2SFCA because it counts populations that fall within the overlap of multiple catchments in step 1 multiple times (e.g., in the example in Fig. 1 the person residing at location 9 will be fully counted in the demand for both R_{s_1} and R_{s_2}), whereas in reality a person's demand for a health service at a specific location is likely to decrease when more spatial alternatives become available. As a consequence, local demand will be overestimated in the 2SFCA method, resulting in biased provider-to-population ratios. To solve this, Wan et al. (2012) have introduced an additional step preceding the 2SFCA procedure in which they split up the demand of a zonal population over multiple health care providers proportional to the distance between the considered zonal centroids and these providers. A drawback of their three-step procedure is, however, that it requires knowledge about the extent to which the selection of medical services depends on distance and how this distance effect is mediated by such factors as health care quality, awareness of health care availability and a patient's health insurance. Furthermore, accessibility values resulting from the procedure are difficult to trace back to any real-world setting, which compromises interpretability. Finally, the demand for health care is generally modeled in step 1 using a single representative spatial proxy point (usually the zonal centroid) or at best by considering the demand to be evenly distributed across the study area. While beneficial in terms of computational tractability, simplified population mappings introduce an additional unnecessary source of spatial bias that is unfavorable for certain health applications where accurate pinpointing of patient locations is essential such as in the case of mapping response times of emergency medical services.

A last method that has been used to model accessibility of health care services is kernel density estimation (KDE). KDE is a non-parametric technique that allows estimating the probability density function of a random variable. In planar KDE, the purpose is to generate a smooth density surface from point events using the following expression (Silverman, 1986):

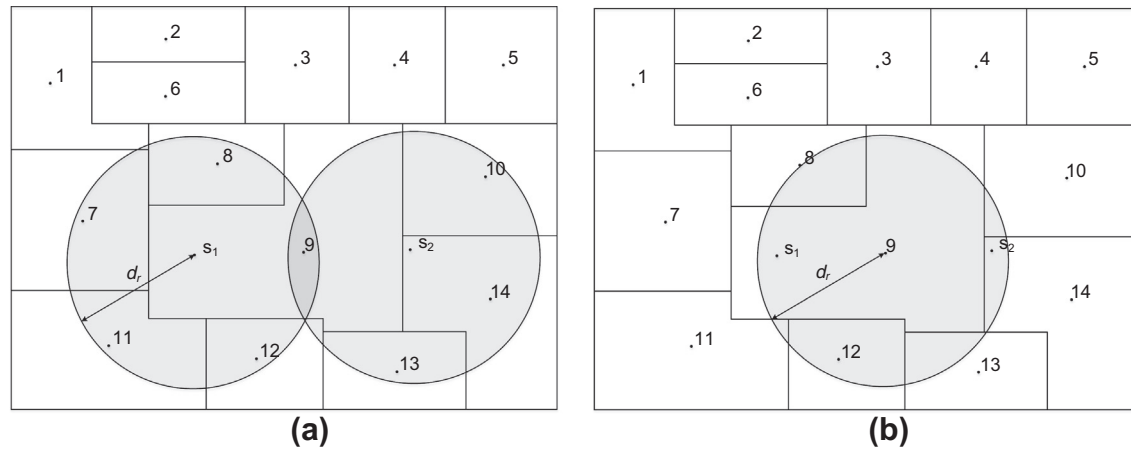


Fig. 1. Floating catchment area method: step 1 (a) and step 2 (b).

$$\hat{\lambda}(s) = \sum_{i=1}^n \frac{1}{\pi r^2} k\left(\frac{s-s_i}{r}\right)$$

where $\hat{\lambda}(s)$ is the estimated density at location s , r is the search radius or bandwidth (only points within r are used to estimate $\lambda(s)$), $k(\cdot)$ is a kernel weighting function and s_i is the location of the observed events. Although it is theoretically possible to compute densities in each and every point of a study area, in practice densities are generally calculated for raster representations of continuous surfaces for the sake of computational manageability. Various functional forms for $k(\cdot)$ have been proposed to weigh an observed point event i at location s_i within the bandwidth according to the distance between the event and the center of the kernel s . Often a radially symmetric Gaussian or quartic function is employed for this purpose. The selection of an appropriate bandwidth represents a critical issue in the KDE approach and is subject to arbitrariness. Large kernel bandwidths will result in more smoothed patterns, whereas small bandwidths will emphasize local peaks and troughs. It is generally accepted that the bandwidth size more strongly affects the results than the choice of a kernel function (Gatrell et al., 1996).

While the KDE method has been extensively exploited for creating disease incidence surfaces and risk surfaces of all sorts in epidemiological studies, its application to model health care accessibility is much less customary. To model healthcare accessibility using KDE, two kernel density raster surfaces of the same resolution and extent have to be combined: one for health care supply with s_i representing the location of health care providers and one for health care demand with s_i representing the demand centroids. The former is then divided by the latter to obtain a spatially continuous equivalent of zonal provider-to-population ratios. While KDE-based accessibility fields yield more detailed spatial patterns convenient for visualization purposes than discrete ratios, there is also a flipside to the smoothing out of point pattern data. One issue is that the neat circular form of kernels used in health studies ignores the influence of transportation networks. Another is that kernel estimates will be distorted near the boundaries of the study area as a consequence of bandwidths stretching out beyond the study area. Both problems can, however, be mitigated to a certain extent by using network-based kernels (Okabe et al., 2009) and incorporating an edge correction term into the kernel model (Jones, 1993). Another item of caution relates to the fact that unoccupied land such as a large lake or forest may distort the kernel estimates of health service density. Important distortions also arise for the population density estimates as soon as the population of a zone is not evenly distributed around the zonal

centroid, since the KDE method assumes the population to be concentrated in the centroid of the zone.

In short, health researchers have used several types of metrics (i.e. minimum travel time/distance, PPR, gravity, expansion method, FCA and KDE) that are conventional in the transport literature to quantify accessibility and made incremental modifications to them to make them suitable for application in health care research. Some of the methods discussed in this section are conceptually more appealing than others but all of them suffer from one or multiple problems. Accessibility metrics have been implemented in GIS in different ways depending on parameter choice and assumptions made. The next section will take stock of the recent empirical contributions that have used GIS to implement one of the accessibility metrics discussed above in the context of health care.

3. Empirical accomplishments

In order to gain insight into the relevant recent accomplishments, a systematic literature search was conducted in May 2014 on three search engines: Web of Science (<http://www.isiknowledge.com>), Scopus (<http://www.scopus.com>) and Pubmed (<http://www.ncbi.nlm.nih.gov/pubmed>). Web of Science is an online citation indexing service maintained by Thomson Reuters that provides access to a multi-disciplinary bibliographic database of *inter alia* 12,000 high impact journals in the technical, medical, and social sciences (including arts and humanities). Scopus is owned by Elsevier and has a larger coverage than Web of Science. It claims to be the largest abstract and citation database with over more than 20,000 peer-reviewed journals. PubMed is a bibliographic database that specifically focuses on references and abstracts on life sciences and biomedical topics. It is maintained by the United States National Library of Medicine (NLM) at the National Institutes of Health (NIH).

Four criteria were set out for the literature selection: (1) the work is published in an international peer-reviewed journal; (2) it is about accessibility in the context of health care; (3) it uses real-world data; and (4) it uses GIS to operationalize accessibility. After some experimentation with several keyword-wildcard combinations and operators, the following search key was defined to find papers that closely match the four criteria: '*care and GIS and accessibility'. The use of the asterisk before "care" enabled retrieving articles with "health care", "healthcare", "primary care", "emergency care", "hospital care" and so on in their title, keywords or abstract. The total number of records found in the three

bibliographic databases (Web of Science, Scopus and Pubmed) was 490 (139, 232 and 119, respectively). After removing 194 duplicates, the number of records was reduced to 296. Each of the three databases produced unique records, which underlines the necessity to consult multiple databases. Next, an initial screening of the articles was performed based on title and abstract and those papers that were patently out of the scope of this literature review (e.g., papers focusing on natural hazards, walkability or food deserts) were deleted so that the number of papers was brought down further to 177. The remaining articles cover a wide range of applications in the field of health care accessibility. To obtain a general view of scholarly attention to the topic, Fig. 2 shows the number of articles published per year for the remaining selection. Given that the research key explicitly required “GIS” to be in the title, keywords or abstract of the paper, it comes to no surprise that no articles were published on the subject before 1991. In the first decade after 1991, there was barely published output. Between 2002 and 2008, there was a moderate and fluctuating productivity, which was followed by a notable rise in publications on the topic between 2008 and 2012. Published output dropped again in 2013, which may constitute the beginning of a downward trend. As shown in Table 1, most of the papers were published in journals with an explicit health focus including *Health and Place*, *Social Science and Medicine*, *International Journal of Health Geographics* and *Tropical Medicine and International Health*, but several papers have also been published in well-known geography journals, including *Applied Geography*. Despite the transport and GIS aspects inherent to the measurement of accessibility, none of the 177 papers were published in a transport-related journal and only few were published in GIScience journals.

In a subsequent step, the number of papers was further reduced to 63 by adding two new selection criteria: (5) the work is about accessibility to primary care, hospital care and EMS (limitation of scope; see introduction) and (6) measuring accessibility to health care services is the main focus of the paper. In other words, papers about specialized care (e.g., dental X-ray services and palliative care) were removed from the selection as well as papers that merely used a GIS-based accessibility metric – often a rudimentary one – as an explanatory variable for differences in health outcome, health behavior or health care utilization. For the remaining coherent collection of articles, the research method to measure accessibility was identified. Fig. 3 shows how often each type of accessibility measure discussed in the previous section has been used in prior work related to primary, hospital and emergency care. If multiple metrics were used in a single paper, the most important or advanced one has been selected. More than a third of the selected empirical papers on healthcare accessibility uses a specific form of floating catchment area (FCA) that is based on the widely cited work by Luo and Wang (2003), which in turn derives from the work by Radke and Mu (2000) on spatial decomposition and on Joseph and Bantock's (1982) gravity formulation.

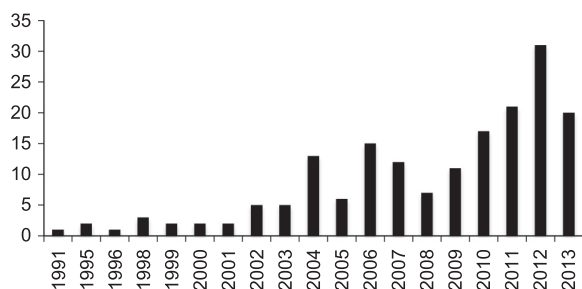


Fig. 2. Number of articles about accessibility to health care per year per year.

Table 1

Most important journals in the field of accessibility to health care.

Journal	No. of articles
Health & Place	16
Social Science & Medicine	11
International Journal of Health Geographics	9
Applied Geography	7
Tropical Medicine and International Health	6

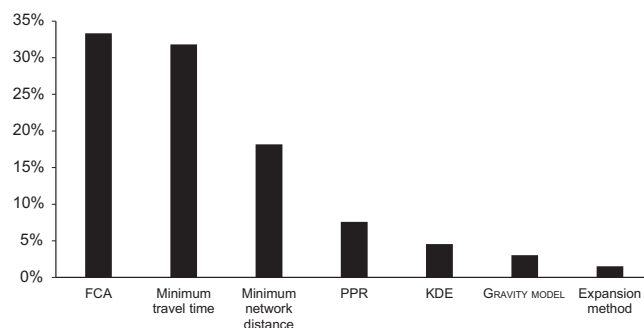


Fig. 3. Frequency in the use of accessibility metrics in the context of primary care.

Despite – or thanks to – their simplicity, travel time and road network distance to the nearest health care service are still popular indicators, even in articles where the main focus is on methodology. This is particularly the case for studies conducted in developing countries where physical separation rather than competition is the key resource constraint in equity discussions (see e.g., Blanford et al., 2012; Huerta Munoz and Källestål, 2012; Owen et al., 2010; Perry and Gesler, 2000; Rosero-Bixby, 2004). In contrast to cognate studies in developed countries, these studies (Blanford et al., 2012; Huerta Munoz and Källestål, 2012; Owen et al., 2010; Perry and Gesler, 2000; Rosero-Bixby, 2004) very often require creative GIS methodologies to deal with the lack of detailed cartographic data. Travel time and distance estimations are also vital in emergency care, as potential EMS response time is known to correlate strongly with morbidity (see e.g., Blackwell and Kaufman, 2002). Hence, a strategic spatial deployment of ground EMS vehicles for advanced life support (ALS) based on response times is necessary to ensure adequate and equitable coverage of pre-hospital emergency care. Another type of method that is used in about 8% of the selected papers is PPR. The use of this method is often framed into a policy discourse on how to deal with health profession shortages (see for instance the work by Juarez et al. (2002) in *Journal of Health and Social Policy*). We found only a few examples of KDE applications – cross-referring one another – in our selection, including Guagliardo et al. (2004), McLafferty and Grady (2004), and Yang et al. (2006). While very often used in transport geography, traditional gravity indicators, including cumulative opportunities, are not so commonly applied in healthcare research, presumably because spatial choice of alternative providers is not valued as such in most health settings. Finally, the expansion method by Paez and colleagues has not (yet) been picked up by other researchers in the health arena and it remains to be seen if it will be in the near future given the data requirements and technicality of the metric.

From this set of closely related papers, a manageable selection of 15 papers was made for more in-depth discussion. Selection was done by the author of this review and based merely on the papers' unique methodological contribution to the field – that is, those papers that introduce new methodologies or examine specific methodological problems were selected. Together, the selected studies provide a synoptic view of the advancements

made in recent years and expose a wide number of salient features regarding assumptions, design and application.

The most important features of the selected papers are summarized in Table 2. The table shows a wide variety in the spatial resolution at which supply and demand are analyzed. On the supply side, service locations are generally geocoded at the postal code or address level, or measured by GPS devices during fieldwork in case studies in the developing world (e.g., Spencer and Angeles, 2007). Demand is much less accurately modeled in spatial terms. Generally, static night-time populations are assumed to be concentrated in a single point within a ward, tract, ZIP code zone or a similar polygonal aggregate. To simplify areas into points, researchers have used both geographical centroids (i.e. mean *xy*-coordinates of all vertices of the boundary of the considered zone) and population-weighted centroids (i.e. mean *xy*-coordinates of the centroids of smaller spatial units such as census blocks within the considered encompassing zone such as a census tract, weighted by their population). There are two exceptions in our list that have used multiple demand representations to specifically explore the ensuing aggregation errors: Langford and Higgs (2006) and Apparicio et al. (2008). Langford and Higgs examined three population representations: (i) population-weighted centroids, (ii) evenly distributed population within a zone, and (iii) dasymetrically distributed population within a zone. The dasymetric method adopted by Langford and Higgs uses a binary mask, which can be extracted from satellite imagery or land-use raster maps, to separate occupied from unoccupied space. They then simply redistribute the zonal population to the occupied space by means of areal, pycnophylactic (i.e. volume-preserving) interpolation. Their model does not, however, distinguish between housing and non-residential (e.g., commercial) buildings. Their results point to important spatial differences in FCA-based accessibility outcomes when different population representations are used. In a similar vein, Apparicio and his co-workers have used both geographical and population-weighted centroids to illustrate the importance of heeding the aggregation error at the demand side. The population-weighted centroids in their study were derived for two sub-zonation schemes of a different spatial resolution (i.e. dissemination areas and census blocks). They concluded that aggregation error particularly affects those census tracts where population density is low and land use predominantly non-residential.

In terms of the attributes related to size and quality that have been associated with health service supply, all studies consider the cardinality of the available set of services (indicated by 'count' in Table 2) but only few have additionally considered other attributes such as staffing capacity (Spencer and Angeles, 2007) and number of hospital beds (Mao and Nekorchuk, 2013). On the demand side, scholars have routinely relied on population size as proxy for health care need without adjustment for socio-economic and other attributes. However, some authors were able to indirectly bring such attributes into the equation. McLafferty and Grady (2004), for example, have examined the association of average clinic density measured by KDE with gender, age, education, ethnicity, marital status and health insurance plan using logistic regression. Paez et al. (2010), for their part, have incorporated various personal and mobility variables in their profile-based estimation of average travel distance.

As mentioned previously, determining an appropriate catchment size represents another critical assumption in most health care accessibility models. Catchment size reflects the importance of a health service and depends among others on the type of care, transport mode and study area. Nevertheless, several studies have simply adopted the 30 min driving time assumption initially put forward by Luo and Wang (2003) or have arbitrarily defined a threshold themselves. Some have explored the sensitivity to catchment size by adopting multiple threshold values and eventually

selected the threshold that seemed 'most plausible'. Tanser et al. (2006) have used Thiessen polygons to delimit catchment areas. Such polygons are entirely based on Euclidean distance and do not account for transportation infrastructure around the health care service. Studies that use a gravity model or a simple distance metric usually do not cut-off the search radius beyond an ultimate travel time or distance implying that all health service are relevant no matter how far away they are located from the demand.

Within the catchment area, the decaying relationship between accessibility and physical separation between potential patients and the health care service is modeled in various ways. Some authors refer to prior work to justify their choice (e.g., Mao and Nekorchuk (2013) and Dewulf et al. (2013)) but the problem is that often the work referred to also did not calibrate the decay properly and applies to an entirely different study area. In our selection, only Haynes et al. (2003) have used observed health care utilization data to empirically calibrate the decay function. In absence of such data, many studies have presumed a rectangular decay function introducing a binary distinction between opportunities or people within and those outside a catchment area. This is particularly so for studies which rely on shortest travel time/distance and FCAs. For gravity models, a linear or inverse power function is often opted for, whereas Gaussian or quartic functions are typically used in KDE methods.

In terms of the design of the selected studies, a first aspect concerns whether or not competition for health services has been taken into account. This question depends on the type of care (services with vs. without significant capacity restrictions) and ties in with the method used in the study. Methods that formulate accessibility in terms of supply–demand ratios (e.g., PPR and FCA) imply that patients have to compete for health services. Individuals residing within an area with high demand and high supply are then considered to experience a comparable level of accessibility as those in low demand and low supply areas, as it yields a similar ratio. A second aspect of design is the way in which transport is modeled in a GIS. Except for Tanser et al. (2006) who have used the raster-oriented Idrisi software, studies have predominantly turned to ESRI® desktop products to estimate travel time/distance with the aid of network analysis. Level of detail of the estimation varies greatly, however; some have carefully modeled link-based speed limitations, one-way streets, traffic sign penalties and so on (e.g., Schuurman et al. (2006)), while others (e.g., Luo and Wang (2003)) have used fixed speed values differentiated by road type and density. None of the selected studies has accounted for congestion and only two studies, Haynes et al. (2003) and Mao and Nekorchuk (2013), have dealt with public transport besides car travel. However, in both studies public transport is modeled rather poorly: while Haynes and others measured access to the bus stop rather than by bus and Mao and Nekorchuk assumed a uniform bus speed within the entire road network. An interesting methodological study with respect to travel time estimation is the work by Delamater et al. (2012) who reviewed the practical differences between raster and network models and the sensitivity of accessibility models to changes in travel speed settings. They demonstrated that GIS design does have a substantial impact on the number of people identified as having limited accessibility, with a raster-based approach being more sensitive to travel speed settings.

The last three columns of the table provide information about the characteristics of the application, including type of care, study area and scale, and key findings of the paper. Three specific key findings deserve more attention here as they further contribute to our understanding of the spatial and GIS-induced errors in health care access studies. The first is Bell et al. (2012) who have zoomed in on geocoding effects. They tested six geocoding methods (e.g., automated, semi-automated, geocoding with areal units,

Table 2

Overview of selected papers representative for the recent advancements in the area GIS-based analysis of accessibility of health care services.

Author(s), year	Method(s)	Assumptions				Design			Application		
		Supply	Demand	Catchment size	Decay	Competition	Transport	GIS software	Type of care	Study area and scale	Key findings
		• Resolution • Attributes	• Resolution • Attributes				• Travel time/distance • Mode				
Luo and Wang (2003)	Gravity	• Census tract level	• Population-weighted centroids per ZIP-code area	∞	Inverse power	No	• Travel speed based on population density and road type • Car	ArcGIS	Primary care physicians	Chicago ten-county region, US; 901 census tracts; 8.3 million people	Gravity measures overestimate accessibility in poor-access areas
	FCA	• Count	• Count	20, 25, 30, 35, 40, 45, 50 min driving time	Rectangular	Yes					
Haynes et al. (2003)	Gravity	• Postal code locations	• Wards and parishes	∞	Linear	No	• Travel speed based on road type • Car and public transport	Arc/Info	GPs	Eastern England; 1373 zones; 19,202 GPs; 2.1 million people	Patient registration intensity decreases linearly with travel time
McLafferty and Grady (2004)	KDE	• Count • Address locations • Count	• Count • Census tracts • Count, socio-economic attributes, insurance plan	2 mile Euclidean radius	Quartic kernel function	No	• Euclidean • None	ArcView 3.2	Prenatal clinics	Brooklyn, US; 144 clinics; 39,000 mothers	Clinic density is positively associated with early initiation of prenatal care
Langford and Higgs (2006)	FCA	• Postal code locations • Count	• Population-weighted centroid; evenly distributed; dasymetrically distributed • Count	10 min driving time	Rectangular	Yes	• Grid cell travel speed based on population density and road type • Car	ArcGIS 9.0	GPs	3 regions in Wales; 2100 GPs distributed in 485 practices	Accessibility depends on alternative spatial representations of population
Schuurman et al. (2006)	Minimum travel time	• Address locations • Count	• Census block level • Count	1-h driving time	Rectangular	No	• Link-based travel speed • Car	ArcGIS 9.1	Hospitals	Southern British Columbia, Canada; 22 hospitals; 655,000 people	Scenarios of service groupings affects population coverage
Tanser et al. (2006)	Minimum travel time	• Address locations • Count	• GPS locations of homesteads • Count	Thiessen polygons	Linear	No	• Raster-based cost allocation • Grid cell travel speed based on road type • Walking and public transport (no timetables, frequencies or waiting times incorporated) • Link-based travel speed • Car	Idrisi Kilimanjaro	Hospitals and clinics	KwaZulu-Natal, South Africa; 1 hospital and 13 satellite clinics; 250 homesteads	Travel time is key in measuring physical accessibility to clinics in developing countries
Yang et al. (2006)	FCA	• Address locations • Count	• Population-weighted centroid of census tract • Count	30 min driving time	Rectangular	Yes		Arc/Info	Renal dialysis centers	Cook county (including Chicago), US; 71 dialysis centers; 0.12% of population; 30 m ² cells	There is a significant mismatch of the accessibility ratios between FCA and KDE

	KDE			1 km around patient residence; 20 km around service	Gaussian kernel function			ArcGIS			
Spencer and Angeles (2007)	KDE	- GPS coordinates of health services - Count, staffing	- Community level centroids - Count	5, 10 and 15 km	Gaussian kernel function	Yes	- Euclidean distance - None	Arc/Info	Physicians, nurses and auxiliary nurses	Nicaragua; 1 km ² cells	Kernel density is a useful method for measuring accessibility to health services in developing countries
Apparicio et al. (2008)	Minimum distance	- Address locations - Count	- Geographical centroid and population-weighted centroid - Count	∞	Linear	No	- Euclidean, Manhattan and network distance - None	ArcGIS	Various types of health services	Montreal census metropolitan area, Canada; 642 health services; 3.4 million people	Aggregation schemes and distance type affect accessibility measurement
Paez et al. (2010a)	Gravity			∞	Inverse power						
	Expansion method	- Address locations - Count	1 km² cells - None	Catchment based on average trip length	Rectangular	No	- Travel speed not a priori assumed but implicit in average trip length - Idem for mode	Not mentioned	Doctors of medicine and dentists	Montreal Island, Canada; 2593 medical and dental offices; 7.6 million	Disparities in health care accessibility depend on age (senior/non-senior), residence location (urban/suburban) and vehicle ownership
Delamater et al. (2012)	Minimum travel time	- Address locations - Count	- Zip code level centroids - Count	30 min driving time	Rectangular	No	- Network-based travel speed based on road type and ownership through decision tree as well as raster-based travel by selecting lowest travel speed within cell - Car	ArcGIS	Hospitals	Michigan, US; 137 hospitals; 9.8 million people	Size of underserved population varied substantially depending on travel speed settings and GIS assumptions (raster/vector)
Wan et al. (2012)	FCA	- Address locations - Count	- Population-weighted census tract centroids - Count	60 min driving time	Gaussian	Yes	- Network-based travel speed based on road type - Car	ArcGIS 9.3	Primary care physicians	Austin-San Antonio corridor, Texas, US; 2.8 million people; 2024 physicians in 855 practices	2SFCA methods overestimate healthcare demand
Bell et al. (2012)	FCA	- Address locations, postal code locations (6 geocoding techniques) - Count	- Dissemination area centroids - Count	3 km	Rectangular	Yes	- Euclidean distance - None	ArcGIS 10	Primary care physicians	Saskatoon, Edmonton and Mississauga, Canada 1995 physicians in 556 practices	Geocoding errors do not significantly affects accessibility values
Dewulf et al. (2013)	Minimum distance	- Address locations - Count	- Census tract centroid - Count	∞	Linear	No	- Network distance - None	ArcGIS 9.3	GPs	Belgium; 10,353 GPs; 10.8 million	Policy makers need to ascertain to what extent there policy evaluations hold under different scales of analysis and when different methods are used
	PPR			Zonal boundaries	None	Yes	- None - None				
	Gravity			Zonal boundaries	Rectangular	No	- None - Network distance - None				
	FCA			5 and 10 km	Stepwise decay	Yes	- Network distance - None				
Mao and Nekorchuk	FCA	Address	Population-	30 min by	Rectangular	Yes	Travel speed based on road type	Not	Hospitals	Florida, US; 365	Accounting for transport mode

(continued on next page)

Table 2 (continued)

Author(s), year	Method(s)	Assumptions	Design			Application					
		Supply	Demand	Catchment size	Decay	Competition	Transport	GIS software	Type of care	Study area and scale	Key findings
		• Resolution • Attributes	• Resolution • Attributes				• Travel time/distance • Mode				
(2013)		locations • Count, number of beds	weighted centroids of census block groups • Count	car/bus			for car, constant bus speed (10 km/h) • Car and public transport	mentioned		hospitals; 11,442 block groups	availability offers better guidance for policy makers to mitigate health inequity issues

(2013)

etc.) and evaluated their impact on health care accessibility assessments. While the six methods exhibited varying levels of positional accuracy, statistically significant effects of geocoding errors on FCA derived accessibility values could not be detected. Also important are the findings by Dewulf et al. (2013). They have highlighted the far-reaching financial and policy implications of inaccurately quantifying health care accessibility. Their study is about the Belgian financial support plan for new GPs who settle themselves in a shortage area. This plan was initiated to redress spatial gaps in health care provision. They showed that in about half of the country financial assistance is awarded in already sufficiently served areas, because a crude PPR criterion lies at the basis of the awarding system.

In sum, several types of accessibility measures have been formulated in the context of access to healthcare services. Minimum network distance or travel time is often applied in academic studies, easy to implement and allow gaining insight into accessibility to services for which freedom of choice is not important. They do however not account for the demand for services. The same is true for kernel density methods, which in essence merely interpolate the availability of services (e.g., hospitals) or their capacity (e.g., hospital beds). They highly depend on the selected interpolation function. Population-to-provider ratios are particularly relevant for policy as they distributed across the population. A bottleneck, however, is the delineation of the areas for which PPR are calculated. Usually artificial zoning systems for which population data is available are used. Another drawback is the neglect of cross-border health care seeking behavior. Gravity measures and measures based on the expansion method permit the incorporation of spatial interaction mechanisms but are more complex and require the estimation of parameters. Their application in health care research has been limited to date. Finally, floating catchment measures are increasingly applied in recent years. They combine the several advantages of other methods: control for demand, allow cross-border healthcare seeking behavior, do not require complex parameter estimation procedures and are easy to implement.

4. Research agenda

While important progress has been made in the papers discussed above, there are still a lot of opportunities for transport geographers to take forward the research agenda. Although the ultimate goal of accessibility metrics is to uncover social disparities in geographical access to health care, current indicators and studies largely fall short in this respect because they suffer from two major geographical problems: the modifiable areal unit problem (MAUP) and the uncertain geographical context problem (UGCoP). The former refers to a well-documented source of statistical bias in spatial analysis whereby the results differ when different data aggregation schemes and scales are employed. The latter, of which the nature and sources have only recently been systematically documented by Kwan (2012a,b), relates to the uncertainty about the area, timing and duration in which individuals are exposed to contextual (i.e. social and environmental) influences thought of to affect their health behavior (e.g., physical activity) and outcome (e.g., obesity).

Current approaches largely suffer from the MAUP because, although detailed geo-referenced data sets on health facilities and their users are increasingly available, accessibility continues to be measured at the level of administrative districts, thereby obfuscating micro-level accessibility problems of social groups and neighborhoods (Omer, 2006) and ignoring important sources of social inequality including mobility resources (Paez et al., 2010a), language barriers (Bell et al., 2013; Flores, 2006) and diverging (often gendered) health needs (Ostlin et al., 2006).

Because all persons residing within the same administrative district – socio-economically different as they may be – are assigned equal levels of accessibility, such large-scale aggregate indicators of health care accessibility do not provide a sufficiently robust description of how access to health care is distributed among a population (Kwan and Weber, 2008). Ecological inference fallacies resulting from aggregate health care accessibility metrics may result in distorted mapping of medical deficits and suboptimal health care facility siting.

Existing indicators also suffer from the UGCoP because they are by nature static in the sense that they fail to account for the fact that a people's exposure to health care fluctuates over time as a result of day-to-day variations in time use and mobility patterns. Because individuals' discretionary time budgets are structured around well-established temporal rhythms of mandatory daily routines (e.g. work, education, chauffeuring children, etc.), time to travel to and spend at health care facilities varies substantially during the day, between days and across individuals. Furthermore, operating hours of health and social services vary from day to day and often conflict with individuals' working schedules. Changes to these hours of operation are known to affect people with different personal and household attributes differently (see e.g., Delafontaine et al., 2011; Neutens et al., 2010a,b). Besides time constraints related to the demand for and provision of health care, transportation also brings about important time constraints in the form of *inter alia* traffic congestion and public transit schedules. While of major importance in practice, these contextual, temporal aspects are, as illustrated in the previous section, almost entirely overlooked in current health access research, where the analytical focus has been on static (night-time) representations of the population. Given that accessibility to health care has hitherto been measured in an aggregate and a-temporal manner, the unequal distributional effects of health care provision across various social groups of the population have never been properly studied.

While in recent years the importance of addressing these lingering caveats has been repeatedly aired by both transport geographers (Kwan, 2012a, 2009; Miller, 2007) and health researchers (Cummins et al., 2007; Perchoux et al., 2013; Schaefer-McDaniel et al., 2010b), there is still a tendency to use simplistic accessibility metrics in public health research, in part because such metrics are relatively easy to implement using popular off-the-shelf GIS software. New accessibility concepts and tools that deal with personal, temporal and equity considerations are sorely needed and may enormously improve our understanding of how geographical accessibility to health care facilities is experienced by different demographic segments. Below I set out the priorities and make suggestions for transport geographers to address the shortcomings of previous studies. I argue that future work should work towards person-based and temporally integrated metrics, more sophisticated geocomputational tools and equity indices that quantify the social disparities in accessibility of health care.

4.1. Towards person-based and temporally integrated metrics

In order to more realistically represent the mobility and temporal constraints that individuals face, researchers may profitably draw on recent developments in the field of time geography (Hägerstrand, 1970). While this framework has served as one of the dominant paradigms undergirding activity-based travel demand modeling, its usefulness for deriving detailed person-based activity spaces in health research has not been fully addressed as of yet (Kestens et al., 2012; Kwan, 2012b). To address this lacuna, health researchers may draw on the space–time measures used by among others Kwan and Weber (2008) and Neutens et al. (2010a, 2010b). The strength of these time-geographical methods is that they consider the demand for services to be

dynamic over space–time by controlling for individuals' mobility and space–time constraints within the study area. However appealing the time-geographic framework may be conceptually, two issues are to be addressed prior to its application in health care research. A first issue has to do with determining the appropriate space–time constraints within which individuals can access health care resources. Earlier implementations of space–time measures have derived these constraints retrospectively from information about where and when individuals participated in fixed activities as reported in activity-travel diaries. But the problem is that medical appointments themselves have been considered fixed activities with a high priority and around which other activities are planned later on. It should be clear, however, that the space–time fixity is a relative and contextual concept (Schwanen et al., 2008). If one assesses the accessibility of healthy food vendors, for instance, a planned medical appointment may represent a fixed constraint whereas it may not in the case of job accessibility. This further depends on the type of medical visit: people will probably take a day off for an occasional appointment with a specialist with a long waiting list, but work constraints are likely to prevail when a minor preventive checkup is considered. These considerations of relative flexibility of activities will need careful thought in future health applications of space–time measures.

Second, while detailed time-geographical procedures enable to explore accessibility trends and patterns at the individual level, their applicability becomes compromised when greater generalizability is targeted so as to produce insights into accessibility disparities among populations in larger study areas. By drilling down at the level of individuals, the analyst may lose sight of the mere spatial patterns of unmet demand. Although geography is increasingly evolving from a data-poor towards a data-rich environment (Miller, 2010), spatiotemporal diary-like information for realistically measuring person-based levels of accessibility within an entire population is yet hard to come by. In most large-scale accessibility evaluations, researchers and policy makers have to rely on demographic data organized at an aggregate level. Furthermore, individual-level space–time accessibility measures do not account for competition effects on the demand side, which are highly important for certain types of medical facilities that face capacity constraints, including general practitioners and hospitals. A solution to upscale the time-geographic metrics would be to develop dynamic, temporally-aware versions of the floating catchments that control for commuting behavior, operating hours of health care services and temporal variations in attainable travel velocity. A good starting point for developing such dynamic catchments is the work by Sadahiro (2005) on spatiotemporal concentration of facility distribution and the work by Delafontaine et al. (2012) on spatiotemporal accessibility. Based on their work, it should be possible to analytically describe spatiotemporal generalizations (i.e. windows) of the current static catchments that can be used in the 2SFCA procedure. These spatiotemporal windows have to be centered on the supply locations (e.g., health care services) in the first step of the procedure and the demand locations (e.g., zonal centroids) in the second step. In a simplified two dimensional non-network constrained space–time environment, these windows will have the shape as depicted in Fig. 4 for a service s_i at location x_i open from t_{i1} to t_{i2} . The supply-centered windows (Fig. 4a) collect all spatiotemporal points from where s_i can be reached given a specified maximum travel velocity v . In other words, the spatiotemporal supply window gathers all x, t -points where $|x - x_i| \leq d_r$ and $t_{i1} \leq t + |x - x_i|/v \leq t_{i2}$. Equivalently, spatiotemporal windows as represented in Fig. 4b will float on the zonal centroids. These demand-centered windows gather all space–time points individuals present at x_j between t_{j1} to t_{j2} can reach, given a maximum travel velocity, or in other words where $|x - x_j| \leq d_r$ and $t_{j1} \leq t + |x_j - x|/v \leq t_{j2}$. Now, in order to estimate both spatiotemporal demand for

and spatiotemporal supply of a health service, we need to formulate functions that describe both supply and demand at each space–time point within these windows. These supply/demand functions can take several forms ranging from simply weighing demand/supply size by distance decay to more complex concentration metrics as those proposed in [Sadahiro \(2005\)](#). The supply/demand functions will be integrated over all x,t -points within the spatiotemporal window. This means that in step one and two of the procedure we will integrate demand and supply, respectively over space–time rather than just space in order to account for temporal fluctuations in the demand and supply at a certain location.

Another alternative is to use the generalized time-geographical metrics that have recently been implemented by [Salze et al. \(2011\)](#), [Widener et al. \(2013\)](#) and [Horner and Wood \(2014\)](#) in the context of food deserts. Salze et al. and Widener et al. use data that are readily available for metropolitan areas around the world (i.e. origin–destination matrices of commuting trips), while Horner and Wood use data that is more difficult to get by (i.e. synthetic travel records). The methods proposed by these authors have relied on network analysis available in conventional desktop GIS environments in order to derive space–time prisms within transportation networks.

Finally, making the accessibility measures more sensitive to interpersonal differences can also be achieved by small yet important refinements with respect to assumptions and design. These include the diversification of the demand for care by adjusting population size by age, gender and other relevant factors, a more accurate modeling of public transit, and the empirical calibration of decay functions by mode based on observed travel behavior.

4.2. Need for more sophisticated geocomputational tools

In developing person-based and temporally integrated metrics, GIScientists have an important role to play because the analytical power of GIS holds tremendous value for public health reformers in uncovering and mapping socio-spatial disparities in health care access, and in monitoring the impact of policy initiatives aimed at reducing these. As recently foretold by [Joyce \(2009\)](#), however, “despite GIS having applications in fields as diverse as engineering and anthropology, the potential of GIS has yet to be fully exploited in health settings” (p. 831). Likewise, [Shaw \(2012\)](#) conclude their recent review of the current state of GIS use in health care research with “a call to action to health informatics researchers to stop ignoring a tool and methodology that has such immense potential for improving the health of our communities”. In a recent study on the influence of neighborhood walkability on physical activity behavior, [Van Dyck et al. \(2009\)](#) go as far as to say that “in Europe, and certainly in Belgium, the use of GIS in health research is not common yet” (p. 503). While in recent years much has been done to reinforce the dialogue between the GIScience and health community, it is fair to say that there is as yet only a modest adoption of sophisticated GIS techniques by health researchers and policy makers.

Researchers are encouraged to move beyond the limited flexibility offered by the standard operations within desktop GIS. Bespoke scripts are necessary to automatize procedures that enable the calculation of health care accessibility at detailed spatiotemporal granularities. These procedures should acknowledge that accessibility fluctuates over time. Recently, [Farber et al. \(2014\)](#) have made an important contribution to the development of time-continuous measures of accessibility. More specifically, they have examined the variation in public transit travel time between all pairs of centroids of a particular zoning system within a study area. Transit travel time and accessibility to supermarkets that was associated with it appeared to fluctuate heavily depending on the considered time of the day as a consequence of

availability and timing of public transportation, showing that single timestamp accessibility calculations may not be representative. Their implementation required timetable data of public transport which has become available in the form of General Transit Feed Specification (GTFS) datasets for an increasing number of regions worldwide ([Hadas, 2013](#)), as well as a Python script that enables to iterate GIS network procedures to derive accessibility metrics and store values of these metrics in indexed tables for various origin–destination combinations and timestamps.

Future work can explore efficient database constructs facilitate storage, management, representation and access to the set of locations that are accessible to a person within a given time budget. These constructs could take the form of an accessibility data hypercube. A hypercube facilitates fast access to pre-computed, summarized data and supports online analytical processing (OLAP) operations including roll-up (aggregation), drill-down (disaggregation), slicing and dicing (selection and projection), and pivot (reorientation of multidimensional data view) ([Gray et al., 1997](#); [Song and Miller, 2011](#)). It essentially consists of facts (*in casu* numerical measures of accessibility) and dimensions (*in casu* perspectives or entities according to which accessibility can be organized). In order to populate the hypercube ([Fig. 5](#)), multiple fine-grained data sources need to be combined including data on health-related facilities (i.e., opening hours, quality, capacity, etc.), information about individuals (i.e., observed of planned activity-travel information, mobility resources, personal and household socio-economic attributes, etc.), and information about the transportation networks (i.e. time-dependent origin–destination shortest path travel times between all pairs of network locations within the considered study area). The implementation and computation of such a hypercube that stores person-based levels of accessibility at different locations and times comprises a major research challenge and will require high performance cloud computing infrastructure to increase performance. I nonetheless believe future research on health accessibility should exploit the capacities of such infrastructures to increase spatiotemporal resolution and derive time-specific and contextual accessibility measurements.

4.3. Measuring equity of accessibility to health care services

Once new concepts have been developed and computational issues have been resolved to measure health care access at finer spatiotemporal resolutions, a final avenue for future work will be to specify metrics that permit to quantify equity in accessibility of health care services. While much has been written about health equity, most of the work to date has focused on equality of treatment and/or health outcome rather than on equality of opportunity. It is important to recognize this fundamental difference in focus because those in equal need and with equal opportunities may not utilize those opportunities to the same degree as a consequence of differences in perceptions, prejudices and preferences about health care ([Goddard and Smith, 2001](#); [Oliver and Mossialos, 2004](#)). Furthermore, the scant studies that have concentrated on equality of opportunity to date have primarily looked at non-spatial factors such as quality, financial affordability and information availability, largely bypassing the distributive effects of health care provision in space and time ([Levesque et al., 2013](#)). Despite the prolific strand of literature on the matter, there is as yet little agreement on how to define and quantify equity of health care accessibility. Furthermore, the social implications of (changes to) the spatial (and temporal) organization of health care delivery are hardly ever considered against the backdrop of ethical theories. While the importance of incorporating ethical principles in the accessibility impacts of transport policies has recently risen to prominence in transport geography and regional science ([Geurs and van Wee, 2011](#); [Martens et al., 2012](#); [Thomopoulos et al.,](#)

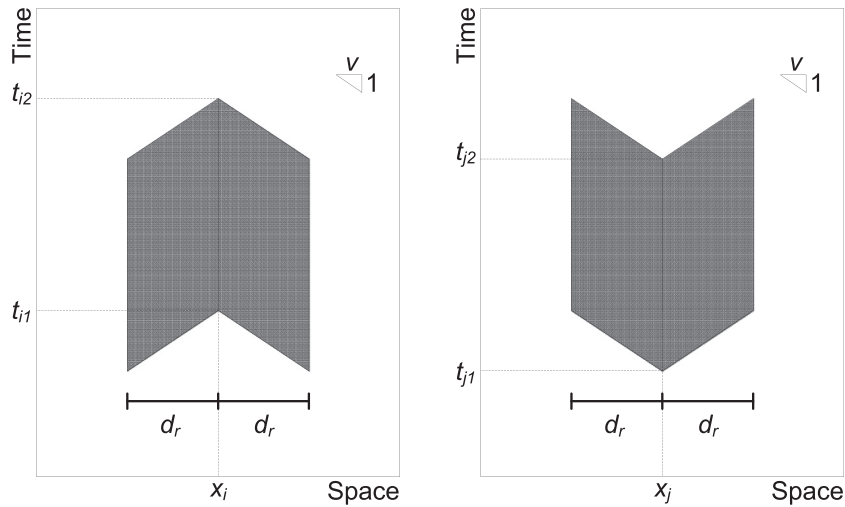


Fig. 4. Spatiotemporal windows: (a) supply-centered window and (b) demand-centered window.

2009), outlining the contours within which equal access unfolds warrants specific systematic treatment within a health context, where equity considerations are pivotal and form part of everyday decision-making.

New work is therefore necessary that empirically quantifies the inequalities in accessibility of health care among persons or groups. One suggestion to achieve this is by modifying existing metrics of income inequality. To this end, researchers may benefit from the work by Waters (2000) and Wagstaff et al. (1991) who discussed different types of equity measures within the context of access to health care including the Gini coefficient and the Utilitarian social welfare function. The Gini coefficient expresses a (weighted) inequality metric that evaluates departure from perfect equality in health care access. The modified Gini coefficient G for measuring (in)equality in health care access can be given by:

$$G = \frac{1}{2n^2\bar{A}} \sum_{i=1}^n \sum_{j=1}^n |A_i - A_j|$$

where A denotes individual-level accessibility, n is the total number of individuals, and $\bar{A}$ is the mean level of A_i over all individuals i . G thus defines equity in access as half of the mean absolute accessibility difference between all individuals, normalized by mean accessibility. The accessibility-specific Gini indicator can alternatively be written as (Barr, 1987):

$$G^* = 1 + \frac{1}{n} - \frac{2}{n^2\bar{A}} (A_1 + 2A_2 + 3A_3 + \dots + iA_i + \dots + nA_n)$$

where group-specific accessibility levels are ranked as $A_1 \geq A_2 \geq A_3 \geq \dots \geq A_i \geq \dots \geq A_n$. This group-specific Gini index G^* introduces weights for each group (e.g., quintiles) depending on the rank that each group has in the overall distribution of a certain variable, with disadvantaged groups at the lower end of the variable spectrum receiving larger weights. In a customary application of an equity metric, one compares the values of a certain variable *in casu* accessibility against the values of that same variable. However, an appealing property of G^* is that it also allows ranking the population groups according to a different distribution so as to contrast the distribution of health care access with, for example, variables of health needs, socio-economic status or level of mobility. The value of G^* ranges from -1 to $+1$, with 0 representing a system where there is no difference in accessibility level across the various population groups (perfect equality), 1 representing a system where the population group with the highest rank (e.g., least needy, richest, most mobile) has access to all the health care and -1 representing a system where the population group with the lowest rank (e.g., neediest, poorest, least mobile) has access to all the health care.

While G^* can facilitate the numerical exploration of ethical viewpoints on equity in accessibility to health care, it disregards the maximization of overall welfare. In other words, an undesirable theoretical situation in which no one has access to the available health care will still be considered by G^* and K as perfectly equitable in egalitarian terms. Hence, we need another metric that should be considered alongside G^* and which typifies the utilitarian perspective. This metric can take the form of the following simple weighted Utilitarian social welfare function:

$$U = \sum_{i=1}^n w_i A_i$$

where w_i ($0 \leq w_i \leq 1$) are weights which are inversely proportional to ranking by health needs, socio-economic status or mobility level. The range of U depends on how accessibility is being defined in the case study at hand, with 0 being the characteristic value denoting a state where no accessibility is produced by the health care system. The above indicators can, when applied together, generate a balanced overall view on how health care accessibility is distributed among a population. Future work applying such indicators will allow making international comparisons in terms of how health care accessibility is distributed within different societies.

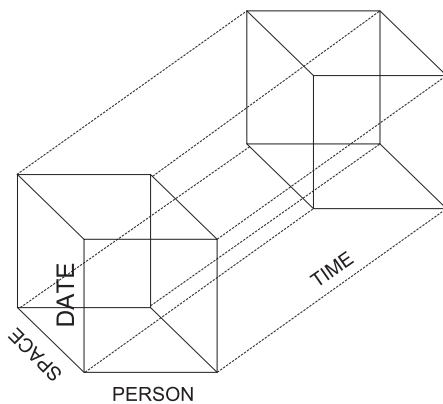


Fig. 5. Accessibility hypercube.

5. Conclusion

This paper has identified various shortcomings in the measurement of accessibility and equity of accessibility of health care services. At the same time, this paper has also offered various suggestions for transport geographers to take forward the research agenda in this field. Exploiting the proposed suggestions will enable to progress beyond the state of the art in respect of two of the most fundamental and widely recognized research challenges in social sciences: the modifiable areal unit problem (MAUP) and the uncertain geographical context problem (UGCoP). More realistically representing the area an individual has access to and is exposed to has not only important ramifications for future studies related to health care accessibility, but also for all kinds of studies that use zonal explanatory variables for individual health and health behavior. The results of ecologically designed health studies have often been inconsistent as a consequence of the use of different contextual units. These units have been conceptualized in stationary and arbitrary zones (e.g., an arbitrary buffer or administrative unit) and therefore do not reflect the areas an individual is truly exposed to (Kwan, 2012b; Perchoux et al., 2013; Schaefer-McDaniel et al., 2010a). This paper therefore invites researchers to develop dynamic and individualized conceptualizations of accessibility for use in, among others, health care accessibility and health exposure assessment. Rather than collecting new data sets on individual travel and health behavior, existing data should be exploited in new ways and with more powerful geocomputational tools in order to obtain a more nuanced picture of how accessibility varies across space–time. Evidently, computational effort should be in proportion to the required generalizability of the results in the context of the research goal and spatial scale at hand.

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